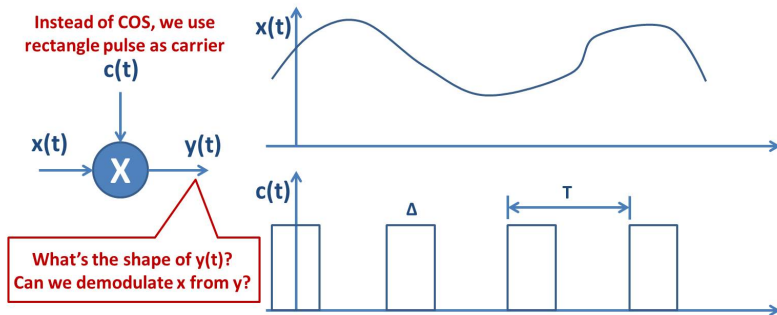


Homework & Tutorial

Homework: 5.39, 5.48 in the attachment

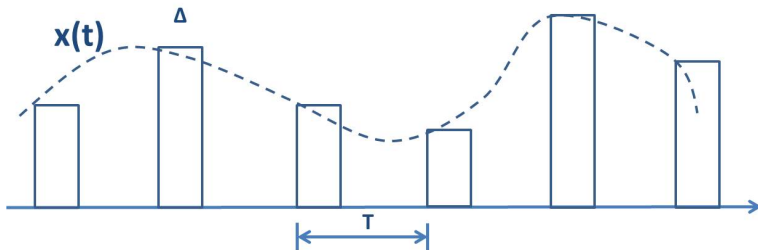
Tutorial: Second homework problem of Chapter 7B, 8.44, 8.46

AM with Pulse-Train Carrier



$$C(j\omega) = \sum_{k=-\infty}^{+\infty} \frac{2\text{sinc}\omega_0 \frac{\Delta}{2}}{k} \delta(\omega - k\omega_0) \quad \text{where} \quad \omega_0 = 2\pi/T \quad (1)$$

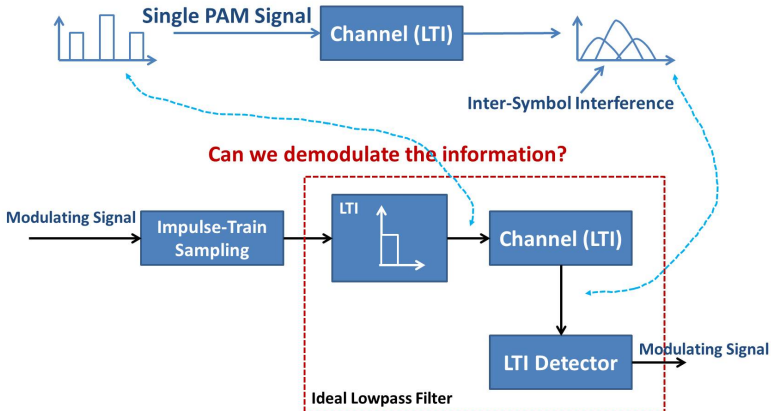
Pulse-Amplitude Modulation (PAM)



- **Pulse-Amplitude Modulation (PAM)**: sample with period T and hold for duration Δ
- Questions:
 - ▶ Without channel distortion, how to demodulate?
 - ▶ Can we use pulse with other shape in PAM?

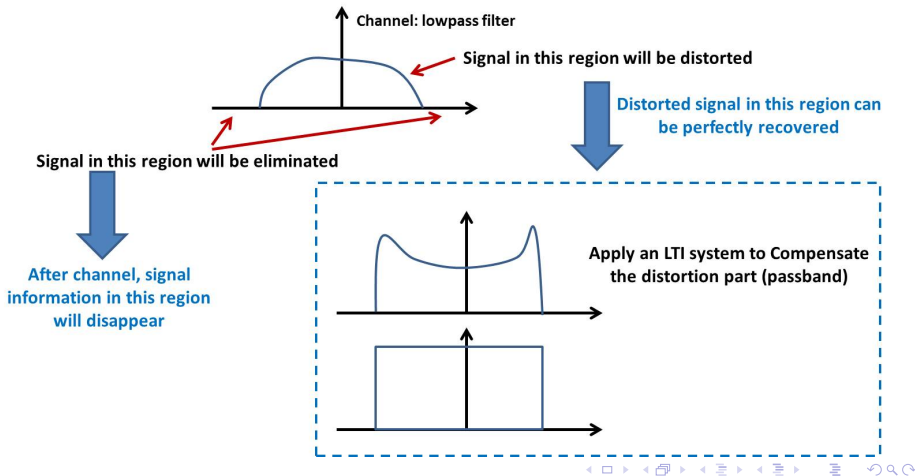
Inter-Symbol Interference (ISI)

- Communication channel is usually a low-pass filter
- Receiver recovers the modulating signal with an LTI system

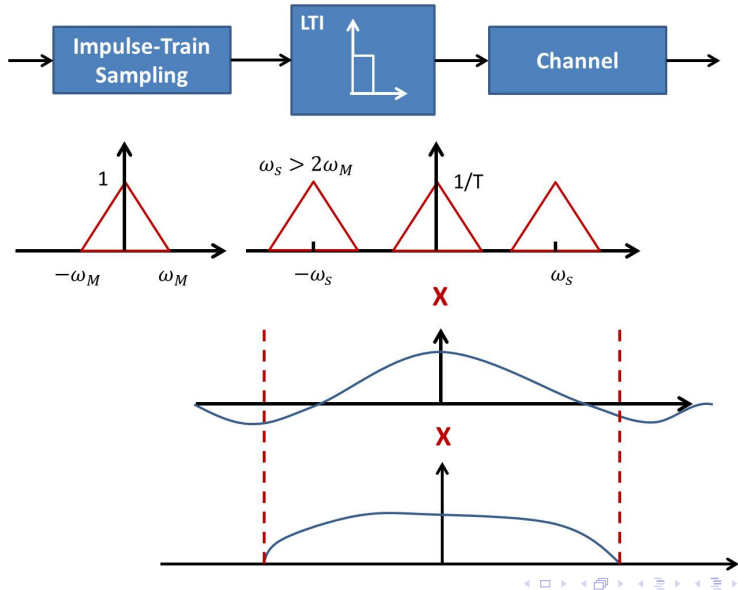


Another Treatment against ISI

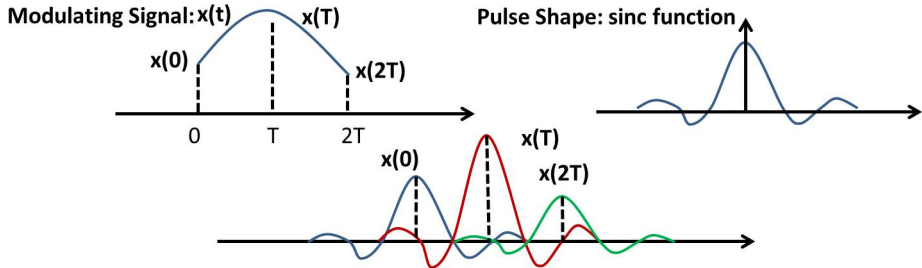
- Another approach: receiver recovers the modulated signal with an LTI system
- What's the cause of ISI?



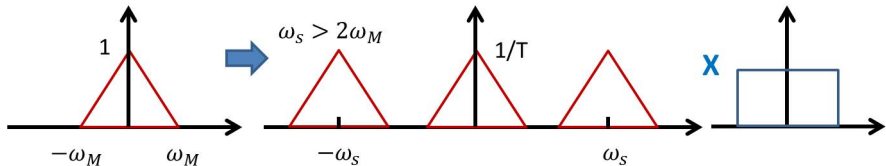
- When using rectangular wave in PAM, the bandwidth is infinity



- To avoid ISI, PAM signal should use band-limited pulse. E.g., sinc function



1. Receiver can perfectly recover this signal
2. Sampled values are kept in the peaks of pulses

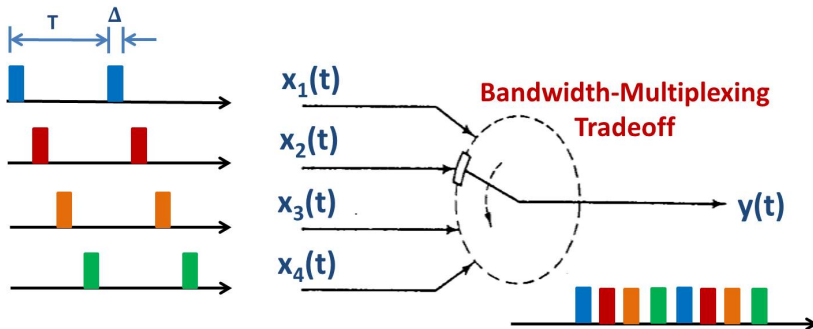


Summary on PAM

- Modulation $\leftrightarrow$ Sampling (zero-order hold); Demodulation $\leftrightarrow$ Recovery
- In addition to rectangular wave, other pulses can be used for PAM
- Demodulation without channel distortion
- Demodulation with channel distortion (ISI)
 - ▶ Recover modulating signal: make sure pulse shaping + channel + receiver's detector = ideal low-pass filter
 - ▶ Recover modulated signal: use band-limited wave in PAM. E.g., sinc wave

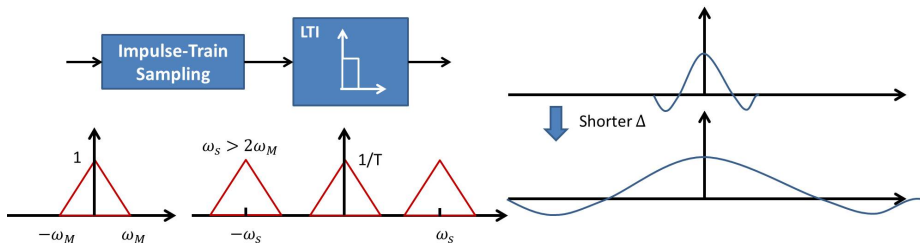
Time-Division Multiplexing

AM signals with pulse-train carrier or PAM signals can be multiplexed in time domain



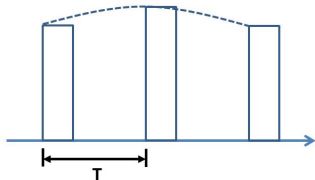
Discussion on TDM

- The number of multiplexed signals is determined by T and Δ
- Can T be as large as we want?
 - ▶ T is the sampling period
 - ▶ Let ω_M be the bandwidth of modulating signal
 - ▶ $\frac{2\pi}{T} > 2\omega_M \Rightarrow T < \frac{\pi}{\omega_M}$
- Can Δ be as short as we want?
 - ▶ Shorter pulse $\Rightarrow$ larger bandwidth consumption



ISI in TDM

PAM Signal 1

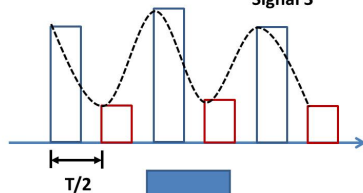


PAM Signal 2

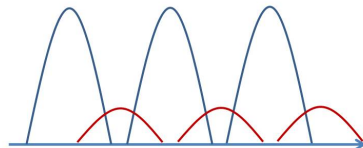


Similar to PAM, we have two approaches to handle ISI

TDM Signal



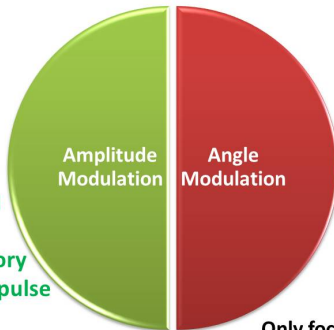
PAM of equivalent Signal 3



What is more ...

Amplitude of carrier contains information

- Sinusoid carrier
 - FDM
- Pulse train carrier
- PAM
 - Demod. without ISI
 - Demod. With ISI
 1. Sampling theory
 2. Band-limited pulse
 - TDM



Angle of carrier contains information

- Phase modulation
- Frequency modulation
 - Narrowband
 - Wideband

Only focus on some basic properties
The demodulator will be introduced in next semester

Angle Modulation

- **Angle Modulation:** Information is carried by the angle of the carrier

$$y(t) = A \cos\left(\underbrace{\omega_c}_{\text{Carrier Frequency}} t + \underbrace{\theta_c(t)}_{\text{Information}} \right)$$

- **Phase Modulation (PM):**

$$\theta_c(t) = \theta_0 + k_p \underbrace{x(t)}_{\text{Modulating Signal}}$$

- **Frequency Modulation (FM):**

$$\frac{d\theta_c(t)}{dt} = k_f x(t) \text{ or } \theta_c(t) = k_f \int_0^t x(t) dt$$

- Instantaneous frequency: $\omega_i(t) = \omega_c + \frac{d\theta_c(t)}{dt}$

Narrowband FM

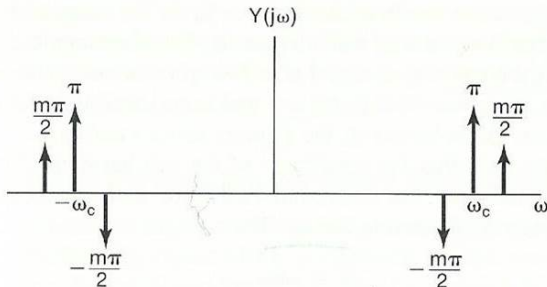
- Modulating signal: $x(t) = As(t)$, where $\max_t |s(t)| = 1$
- **Narrowband condition:** $|\theta_c(t)| = |k_f A \int_0^t s(t) dt| \rightarrow 0$
- Modulated signal:

$$\begin{aligned}y(t) &= \cos\left[\omega_c t + k_f A \int_0^t s(t) dt\right] \\&= \cos\omega_c t \cos\left(k_f A \int_0^t s(t) dt\right) - \sin\omega_c t \sin\left(k_f A \int_0^t s(t) dt\right) \\&\approx \cos\omega_c t - \left(k_f A \int_0^t s(t) dt\right) \sin\omega_c t\end{aligned}$$

- **Observation:** Narrowband FM is similar to AM
- **How is the power efficiency of narrowband FM?**

Narrowband FM Example

- Let $x(t) = A\cos\omega_m t$, $\theta_c(t) = \frac{k_f A}{\omega_m}(\sin\omega_m t)$
- Narrowband Condition: modulation index $m = \frac{k_f A}{\omega_m} \rightarrow 0$
- $y(t) \approx \cos\omega_c t - m(\sin\omega_m t)(\sin\omega_c t)$
- **Observation:** Bandwidth ($2\omega_m$) of narrowband FM signal is determined by the bandwidth of modulating signal, independent of its amplitude



Wideband FM

- Example: $x(t) = A\cos\omega_m t$
- Modulated signal: $y(t) = \cos(\omega_c t + \frac{k_f A}{\omega_m} \sin\omega_m t) = \cos(\omega_c t + m\sin\omega_m t)$
- Using Bessel functions J_n , we have

$$\begin{aligned}\cos(\omega_c t + m\sin\omega_m t) &= J_0(m)\cos\omega_c t \\ &\quad - J_1(m)(\cos(\omega_c - \omega_m)t - \cos(\omega_c + \omega_m)t) \\ &\quad + J_2(m)(\cos(\omega_c - 2\omega_m)t + \cos(\omega_c + 2\omega_m)t) \\ &\quad - \dots \\ &= \sum_{n=-\infty}^{\infty} J_n(m)\cos(\omega_c t + n\omega_m t)\end{aligned}$$



Bandwidth of Wideband FM

m	J0	J1	J2	J3	J4	J5	J6	J7	J8
0	1								
0.25	0.98	0.12							
0.5	0.94	0.24	0.03						
1.0	0.77	0.44	0.11	0.02					
2.0	0.22	0.58	0.35	0.13	0.03				
3.0	-0.26	0.34	0.49	0.31	0.13	0.04	0.01		
4.0	-0.40	-0.07	0.36	0.43	0.28	0.13	0.05	0.02	
5.0	-0.18	-0.33	0.05	0.36	0.39	0.26	0.13	0.05	0.02

- If $m = 0.25$, the bandwidth is $2\omega_m$
- If $m = 0.5$, the bandwidth is $4\omega_m$
- **Carson's rule:** the bandwidth of FM signal is approximated to be $2(m + 1)\omega_m$